

STAT 301 In-class Activity No. 2

Consider the “wage” dataset in Activity No. 1 again. We fitted a simple linear regression model, with WAGE as response and EDUCATION as a predictor, in Activity No. 1. Answer the following questions:

- (a) Is the relationship between WAGE and EDUCATION statistically significant?
- (b) Find a 95% confidence interval for β_1 (slope in the simple linear regression model). Is the result consistent with (a)?
- (c) What is the relationship between the 95% confidence interval for β_1 and the z -statistic (and the corresponding p-value) from R output?